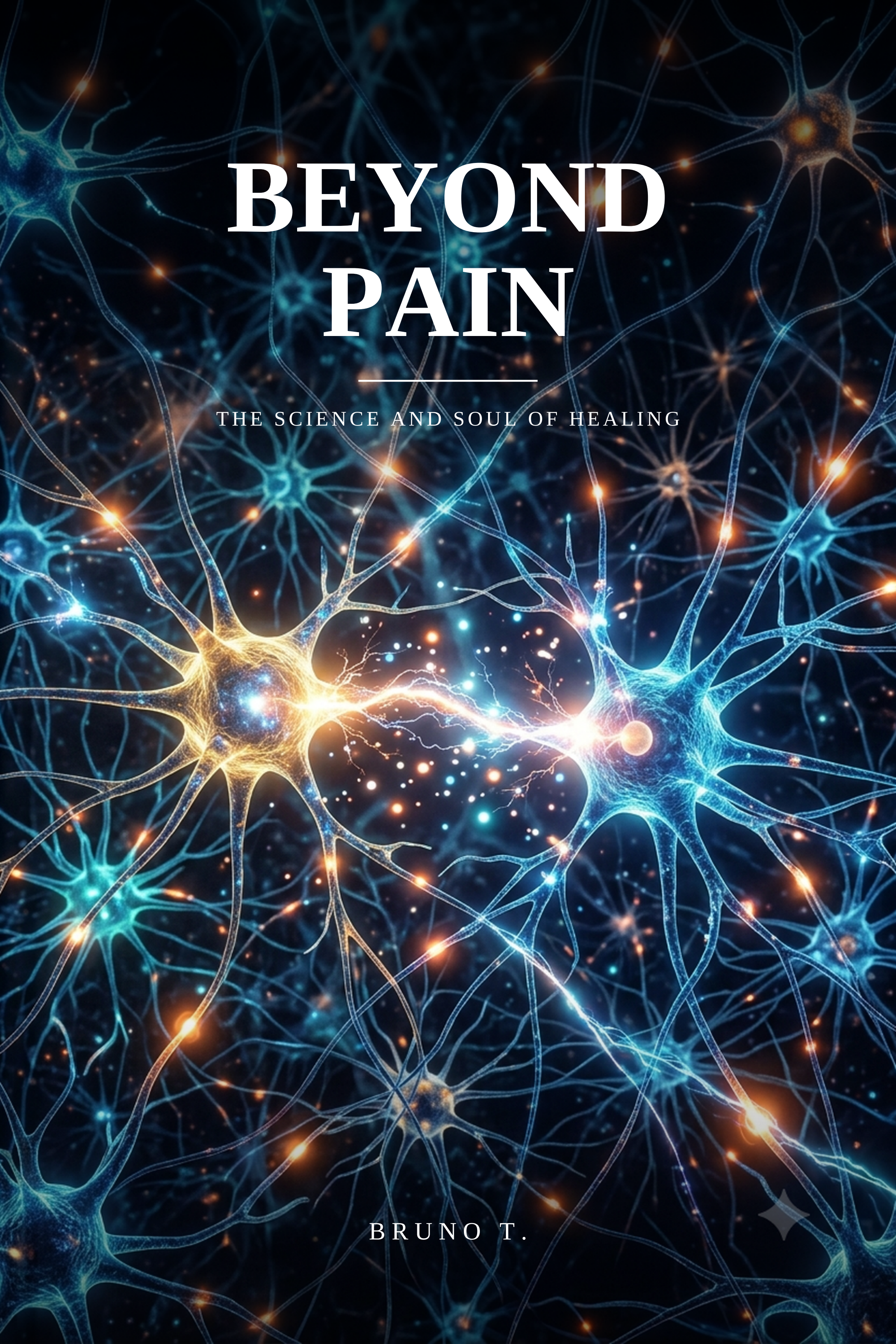

BEYOND PAIN

THE SCIENCE AND SOUL OF HEALING

BRUNO T.

BEYOND PAIN

The science and soul of healing — body, soul and spirit

Bruno T.

BPR Physical Rehabilitation

To my wife, and my rest;
to my children, the daily reason for my purpose;
to my parents and my family, who gave me roots and hold me up.

And to every person who carries a pain and, even so, dares to hope:
this book is for you.

Opening

Pain That Speaks

Let me begin with a confession you might not expect from a book about pain: I don't write this from a distance.

Much of what you'll read here I learned first in my own body, and only later in the books. I know the pain that wakes a person in the small hours, the pain that steals whole plans, the pain that makes the future look smaller than it did yesterday. And I know the other side too — the day when pain, which seemed only to destroy, began to say something. It was when I learned to listen to it that everything changed. That listening is what I want to teach you.

Because pain speaks. It's we who, almost always, don't know its language.

We're taught to treat pain as an enemy: something to be silenced as quickly as possible, with a tablet, a procedure, a distraction. And sometimes silencing it is exactly right. But when pain persists — when it returns, or won't leave, or appears without any scan explaining why — treating it only as an enemy means missing the message it came to bring. Because pain, deep down, is a **messenger**. It isn't the problem itself; it's the alarm pointing to the problem. And there's an enormous difference between switching off an alarm and solving what it's trying to tell you.

This book was born of a simple conviction, which my own story and my work have only confirmed: **a human being is not made of pieces.** Modern medicine, brilliant in so many ways, has learned to divide us — one doctor for the bone, another for the mind, and, if there's time left over, someone for the spirit, in a separate appointment. But you don't live in pieces. You are one. And your pain, almost always, is one too: it moves between body, soul and spirit with a freedom our clinics insist on ignoring. An unresolved stress becomes back pain. An emptiness of meaning becomes a tiredness that sleep can't cure. A poorly cared-for body wears down the soul. To treat one without looking at the others is, in the end, to treat half the person.

That's why I built this book around a very old line, written almost two thousand years ago: *"may your whole spirit, soul and body be kept complete"* (1 Thessalonians 5:23). Three dimensions, one being. That is the map we'll follow, and it gives the shape of the four parts ahead of you. We'll begin with the **body** — because that's where pain shouts first, and because there are wonders there that will change the way you understand every pain you've ever felt. We'll rise to the **soul** — the mind, the emotions, the hurry of the world — and see how what we carry inside settles into the flesh. We'll go down to the deepest place, the **spirit** — the dimension our age has all but forgotten, and where an emptiness lives that nothing else can fill. And we'll end in **integration**: how to care for all three together, in the routine of a real life, towards a purpose.

Right here at the doorway, I need to make you two promises — and one warning.

The first promise is of **scientific honesty**. Every mechanism, every study, every figure I cite is real and referenced at the end of each chapter. I won't sell you miracles, magic frequencies or guaranteed cures. The real science of pain is astonishing enough on its own; it needs no decoration.

The second promise is of **spiritual honesty**. I am a person of faith, and I won't pretend otherwise — the dimension of the spirit carries, for me, a weight this book couldn't hide without betraying itself. But faith, here, will never be used to blame the one who suffers, nor to promise what cannot be delivered. I'll treat it with the same care and the same truth with which I treat the science. And I'll show you something that delights me: how often ancient wisdom and modern discovery, when honest, say the same thing in different languages. That combination — faith and science side by side, without either diminishing the other — is the heart of this book.

The warning is short: **this book is not a substitute for your doctor or your therapist.** It's a map, not a diagnosis. A new pain, one that changes its pattern or comes with warning signs, needs professional assessment — and part of wisdom is knowing when to seek help. What I offer here works *alongside* good medical care, never in its place.

One last word, before we turn the page. If you've come to this book carrying a pain — in the body, in the soul, or in that deep part we can't always name — I want you to know, from the very start, that your pain is real, that it is known, and that your story is not over. I don't promise it will be gone by the last page. I promise something better: that you'll learn to listen to it, to care for yourself as a whole, and to discover that there is life — a great deal of life — beyond it.

Pain is speaking. Let's learn, together, to hear it.

Chapter One

Pain From the Inside

Two workmen. Two nails. Two opposite fates — and, between them, a lesson that turns almost everything we think we know about pain inside out.

The first case was published in one of the world's most respected medical journals, the *British Medical Journal*, in 1995. A 29-year-old builder arrived at A&E after jumping onto a fifteen-centimetre nail, which had gone straight through the sole of his boot. The slightest movement of the nail was unbearable. The pain was so intense that the doctors had to sedate him with fentanyl and midazolam — heavy drugs — before they could even touch the wound. Only then did they remove the nail, drawing it out from below. And when they took off the boot, something that looked like a miracle appeared: the nail had passed **between the toes**. The foot was untouched. Not a scratch, not a drop of blood (Fisher, Hassan and O'Connor, 1995).

Now hold that image, and set the opposite story beside it. A man was working with a nail gun when the tool went off by accident near his face. He felt a knock, a mild toothache, a bruise on his jaw — and got on with life. He ate, slept, worked. Six days later, he went to the dentist about that "toothache". The X-ray revealed a ten-centimetre nail **lodged inside his head**, driven through his skull.

One man in agony over a nail that never touched him. Another almost pain-free with a nail buried in his brain. How is that possible?

The answer is the key to this whole book, so it's worth saying slowly:

Pain is not the damage. Pain is the interpretation your brain makes of the damage.

We grow up believing pain is a faithful report from the body: the more it hurts, the worse the injury. It's intuitive — and it's wrong. Pain is not measured at the injured spot. It is **produced by the brain**, after it weighs a single essential question: *"how much danger am I in?"* In the first workman, the eyes saw a nail through the boot, colleagues shouted, everything said "catastrophe" — and the brain produced a real and devastating pain, with no injury at all. In the second, nothing said "danger" — and the brain stayed quiet, despite the gravest damage.

Look closely: in both cases the pain (or its absence) was **real**. No one was pretending. The point isn't that pain is "made up". The point is that pain is a **decision** by the nervous system about how much you need protecting — not an automatic reading of the harm (Moseley and Butler, 2015).

To understand how that decision is made, let's do what this chapter's title promises: go inside. And in there, pain has a smell, an electricity and a chemistry.

The soup of danger: the chemistry of a pain beginning

Scattered through your skin, muscles, joints and organs are millions of nerve endings called **nociceptors**. The name comes from the Latin *nocere*, "to harm". And here lies the first surprise: they are not "pain receptors". They are **danger detectors**. They don't feel pain — they warn.

What makes a nociceptor fire? Often, pure chemistry. When tissue is injured — a sprain, a burn, a cut — the ruptured cells and the immune system pour out a genuine **chemical cocktail**: prostaglandins, bradykinin, substance P, histamine, hydrogen ions (acidity) and ATP. Scientists nicknamed this broth the "inflammatory soup". It does two things. First, it activates the sentinels. Second — and this is what explains why an injured area becomes sensitive to the lightest touch — it **lowers the threshold** of the nociceptors: it leaves them hair-triggered, ready to fire at almost nothing. That's why the skin around a burn stings at a breath. It isn't your imagination; it's the local chemistry turbo-charging the sensors.

Hold on to that word: **sensitisation**. It will return, in ever deeper forms, throughout the book.

The master switch of pain

When the sentinel decides to fire, it has to turn the threat into electricity — a nerve impulse. And how does a nerve, made of flesh, "produce electricity"? By opening tiny gates in its membrane through which charged sodium particles rush in. That flood of sodium is the electrical impulse itself.

One of those gates has a first and last name: **Nav1.7**, a sodium channel that sits precisely in the danger neurons. And it is so central that it deserves a story.

In the early 2000s, doctors in northern Pakistan were puzzled by a boy who performed in the streets as a kind of pain artist: he ran knives through his own arms and walked over hot coals — feeling absolutely nothing. He wasn't brave; he was **unable to feel pain**. Researchers studied his family and two others and found the cause: a mutation that **switches off the Nav1.7 channel**. Without that channel, the sentinel can't fire, the warning never leaves, and the person lives in a world without pain (Cox et al., 2006). (The story has a tragic and revealing end: the boy is thought to have died leaping from a roof, on his fourteenth birthday. With no pain to warn him, danger had no voice.)

Why does this matter to you, who want to ease your pain? Because if there is a chemical switch for pain, it's possible to **act on it**. The local anaesthetics your dentist injects work exactly this way: they block the sodium channels and the warning never rises. And laboratories the world over are chasing a painkiller that targets Nav1.7 — the dream of a "pain off-switch" without the effects of opioids. The chemistry that causes pain is also the door to silencing it.

Why chilli burns (and what that teaches about relief)

Try a thought experiment: why does a chilli "burn" the mouth if it isn't hot? Because your nociceptors have a heat sensor called **TRPV1** — and capsaicin, the substance in chilli, slots into that sensor and fools it. The brain receives the signal "I'm burning" and produces a genuine burning sensation, with no high temperature at all (Caterina et al., 1997). It was such a fundamental discovery that it earned its author, David Julius, the Nobel Prize in Medicine in 2021.

And here comes the useful twist: that same sensor became a medicine. **Capsaicin** creams — yes, the principle of chilli — are used to treat certain pains. By stimulating TRPV1 in a controlled, repeated way, they deplete the nerve's substance P and, over time, **desensitise** that region. The chemistry that burns, in the right dose and the right way, also calms.

The signal's journey and the gate that controls the volume

Once fired, the danger warning runs along the nerves by two "roads". The fast fibres (A-delta) carry the alert at full speed — that's the sharp, immediate pain that makes you pull your hand from the stove before you even think. And the slow fibres (C fibres) come just behind, more slowly, carrying the dull, throbbing pain that lingers. That's why the pain of a knock tends to arrive in two waves.

The signal climbs to the **spinal cord** and reaches a first checkpoint. It's not a pipe you simply pass through: it's a **gate**. In 1965, two scientists, Ronald Melzack and Patrick Wall, published the **gate control theory** in the journal *Science* (Melzack and Wall, 1965). They explained something everyone does without thinking: when you bang your elbow, you rub the spot — and it helps. Why? Because at that gate the pain signals compete for room with the touch signals. By rubbing, you flood the gate with touch and **muffle** part of the pain warning before it rises.

And the most striking thing: that gate also takes orders **from the top down**, from the brain itself, which can send messengers down to **open or close the gate**. Your brain has, quite literally, a volume dial for pain.

That's why a soldier wounded in combat, or an athlete mid-match, sometimes only notices the injury afterwards: the brain, busy with something more urgent, kept the gate shut.

Only after passing that gate does the signal reach the brain. And up there, there is no single "pain centre". There is a **network** — many regions, including those of emotion, memory and meaning. That network compares the warning with everything you are, and only **then** decides to produce — or not — the pain. So it's worth repeating: **pain is not something the body sends to the brain; it's something the brain builds.**

The pharmacy inside you

There's one more detail most people have never heard: your body makes its own painkillers. When the brain decides to turn down the volume of pain, it releases substances called **endorphins** — chemical cousins of morphine, produced by you yourself. It's part of what explains the "runner's high", the strange relief in the middle of a shock, and even a slice of the placebo effect.

And the best part: this inner pharmacy **can be opened**. Movement, laughter, music, prayer, the touch of someone you love, a purpose that pushes you forward — all of it helps release your body's own opioids. You are not merely a passive victim of pain; you carry, inside you, part of the medicine.

The gift almost no one wants

If pain is such hard work — chemical soup, sodium channels, roads, a gate, a whole courtroom in the brain — the question is fair: what is all this for? Wouldn't a life without pain be better?

Meet Jo Cameron. Scottish, now in her seventies, she lived almost her whole life without knowing she was different. Jo **feels virtually no pain**. She also feels almost no fear or anxiety, and heals quickly. It sounds like a blessing — until you hear the details. Jo has burned her own skin on the stove and only noticed **by the smell of burning flesh**, because pain never warned her. She discovered severe arthritis in her hip only when doctors were baffled that such a destroyed joint didn't hurt. She has come through operations that are usually very painful using little more than an ordinary painkiller. The cause: rare mutations in the FAAH and FAAH-OUT genes, which keep the volume of pain permanently low (Habib et al., 2019; Mikaeili et al., 2023).

Jo's story, and the Pakistani boy's, teaches something our culture, obsessed with escaping discomfort, has forgotten: **pain is a gift.** It's an alarm system that keeps us alive. Those born without it injure themselves without noticing, and tend to die young. The pain you so want to silence is the same one that has already made you pull your hand from the fire and seek help in time. It is not your enemy. It is a faithful sentinel.

So, what to do with pain? (the first solutions)

If you've followed this far, you've already gained the first and most powerful tool. Let's organise the solutions that flow from all this:

1. Understanding pain already reduces it. It sounds too good, but it's science. When a person grasps that pain isn't the same as damage — that the alarm sometimes exaggerates — the brain begins to interpret the signal as less threatening, and the pain actually drops. Pain education programmes reduce pain and disability in measurable ways (Louw et al., 2016; Moseley and Butler, 2015). This chapter, in truth, is already treatment.

2. Close the gate. Melzack and Wall's theory became an everyday tool: rubbing the spot, applying heat or cold, or using a **TENS** machine (which sends electrical touch) helps "close the gate" and muffle the pain. Simple relief, with a real physiological basis.

3. Adjust the chemistry — wisely. Anti-inflammatories like aspirin and ibuprofen work by blocking the production of the prostaglandins in that inflammatory soup — a discovery that earned John Vane the Nobel

Prize in 1982 (Vane, 1971). They have their place, especially in acute pain. But they have limits: they don't resolve the cause, and prolonged use exacts its price on the stomach, the kidneys and the heart. Capsaicin creams and anaesthetic patches (which block sodium) are useful local options for certain pains.

4. Open the inner pharmacy. Movement, sleep, connection and purpose release your own endorphins. It isn't empty "positivity"; it's biochemistry.

5. And the deepest solution of all: give the brain **safety**. Because the brain produces pain from the question "how much danger am I in?", everything that reduces the sense of threat — understanding, moving with confidence, sleeping, calming the mind, having hope — turns down the volume at its source. This is what the rest of the book will deal with, layer by layer.

The body is a marvel — and that isn't just poetry

Pause for a moment and look at what we've described. Sentinels scattered throughout the body. A chemical soup that wakes them. Sodium gates that become electricity. Two roads, one fast and one slow. A gate that opens and closes. A brain that weighs danger, memory, emotion and meaning. And an inner pharmacy of painkillers. All of it in thousandths of a second, without your asking, all the time, to protect you.

Three thousand years ago, a man looked inside himself without any microscope and wrote: *"I praise you, for I am fearfully and wonderfully made"* (Psalm 139:14). He didn't know the names — nociceptor, Nav1.7, endorphin. But he had the right intuition. The body is not a dumb machine that breaks and hurts. It is a work of engineering whose wisdom still leaves us open-mouthed. And the system that makes you suffer is the same one that keeps you safe.

That changes your relationship with your own pain. It stops being a random punishment and returns to what it always was: **a messenger.**

The point is not to hate it, nor merely to silence it, but to learn to listen to what it came to say — because, as you'll see, the message is rarely only about the tissue.

An honest caveat

I need to be clear, so you don't misunderstand me. To say the brain builds pain does **not** mean it's "imagination" or "making a fuss". Pain is always real. And to say pain doesn't always reflect the damage does **not** mean injuries don't matter. A fracture, an infection, certain warning signs still demand real and sometimes urgent care. What changes is this: pain is not a reliable ruler of the harm. It is a protector that sometimes over-sounds the alarm — and sometimes fails to sound when it should. Understanding that doesn't diminish your pain. It frees you to care for it the right way.

What comes next

We've seen how pain is born, travels and is decided — and why, deep down, it exists for your good. But that raises an uncomfortable question, which may be yours: *what about when the alarm simply won't switch off?* When the injury healed long ago, but the pain goes on, month after month, year after year? It isn't the body lying. It's something else — and that's where we're going now.

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Holy Bible, Psalm 139:14.

What’s ahead

In This Book

You’ve just read Chapter One. Here is the whole journey it opens — twelve chapters, in four parts, from the body all the way to a purpose beyond the pain.

This is where Chapter One ends — the first step of *Beyond Pain*.

The book is being written now, chapter by chapter, and you can follow the whole journey. Each new chapter goes first to readers like you.

bpr.rehab/beyond-pain

A note of honesty: this book won't promise miracle cures or quick fixes. It offers something more trustworthy — a clear-eyed look at how pain really works, faith that never blames the one who suffers, and real hope for the whole person. It is a companion to your care, never a replacement for your doctor or therapist.